Neural Computation -- 1998

115 article(s), all issues

1. Information Maximization and Independent Component Analysis: Is There a Difference?

ObradovicD., DecoG.

https://doi.org/10.5555/1122708.1122718

This article provides a detailed and rigorous analysis of the two commonly used methods for redundancy reduction: linear independent component analysis (ICA) posed as a direct minimization of a sui...

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2. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1998.10.issue-1

No abstract available.

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3. (untitled)

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https://doi.org/10.1162/neco.1998.10.issue-4

No abstract available.

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4. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1998.10.issue-6

No abstract available.

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5. (untitled)

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https://doi.org/10.1162/neco.1998.10.issue-3

No abstract available.

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6. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1998.10.issue-5

No abstract available.

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7. (untitled)

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https://doi.org/10.1162/neco.1998.10.issue-2

No abstract available.

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8. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1998.10.issue-8

No abstract available.

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9. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1998.10.issue-7

No abstract available.

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10. Memory Maintenance via Neuronal Regulation

D. Horn, Nir Levy, Eytan Ruppin

https://doi.org/10.1162/089976698300017863 | pp. 1-18

Since their conception half a century ago, Hebbian cell assemblies have become a basic term in the neurosciences, and the idea that learning takes place through synaptic modifications has been accepted as a fundamental paradigm. As synapses undergo continuous metabolic turnover, adopting the stance that memories are engraved in the synaptic matrix raises a fundamental problem: How can memories be maintained for very long time periods? We present a novel solution to this long-standing question, based on biological evidence of neuronal regulation mechanisms that act to maintain neuronal activity. Our mechanism is developed within the framework of a neural model of associative memory. It is operative in conjunction with random activation of the memory system and is able to counterbalance degradation of synaptic weights and normalize the basins of attraction of all memories. Over long time periods, when the variance of the degradation process becomes important, the memory system stabilizes if its synapses are appropriately bounded. Thus, the remnant memory system is obtained by a dynamic process of synaptic selection and growth driven by neuronal regulatory mechanisms. Our model is a specific realization of dynamic stabilization of neural circuitry, which is often assumed to take place during sleep.

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11. Ordering of Self-Organizing Maps in Multidimensional Cases

Guang-Bin Huang, H.A. Babri, Hua-Tian Li
https://doi.org/10.1162/089976698300017872 | pp. 19-23

It has been proved that in one-dimensional cases, the weights of Kohonen's self-organizing maps (SOM) will become ordered with probability 1; once the weights are ordered, they cannot become disordered in future training. It is difficult to analyze Kohonen's SOMs in multidimensional cases; however, it has been conjectured that similar results seem to be obtainable in multidimensional cases. In this note, we show that in multidimensional cases, even though the weights are ordered at some time, it is possible that they become disordered in the future.

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12. Predicting the Distribution of Synaptic Strengths and Cell Firing Correlations in a Self-Organizing, Sequence Prediction Model

Asohan Amarasingham, William B. Levy
https://doi.org/10.1162/089976698300017881 | pp. 25-57

This article investigates the synaptic weight distribution of a self-supervised, sparse, and randomly connected recurrent network inspired by hippocampal region CA3. This network solves nontrivial sequence prediction problems by creating, on a neuron-by-neuron basis, special patterns of cell firing called local context units. These specialized patterns of cell firing--possibly an analog of hippocampal place cells--allow accurate prediction of the statistical distribution of synaptic weights, and this distribution is not at all gaussian. Aside from the majority of synapses that are, at least functionally, lost due to synaptic depression, the distribution is approximately uniform. Unexpectedly, this result is relatively independent of the input environment, and the uniform distribution of synaptic weights can be approximately parameterized based solely on the average activity level. Next, the results are generalized to other cell firing types (frequency codes and stochastic firing) and place cell-like firing distributions. Finally, we note that our predictions concerning the synaptic strength distribution can be extended to the distribution of correlated cell firings. Recent published neurophysiological results are consistent with this extension.

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13. State-Dependent Weights for Neural Associative Memories

Ravi Kothari, Rohit Lotlikar, M. Cahay
https://doi.org/10.1162/089976698300017890 | pp. 59-65

In this article we study the effect of dynamically modifying the weight matrix on the performance of a neural associative memory. The dynamic modification is implemented by adding, at each step, the outer product of the current state, scaled by a suitable constant eta, to the correlation weight matrix. For single-shot synchronous dynamics, we analytically obtain the optimal value of eta. Although knowledge of the noise percentage is required for calculating the optimal value of eta, a fairly good choice of eta can be made even when the amount of noise is not known. Experimental results are provided in support of the analysis. The efficacy of the proposed modification is also experimentally verified for the case of asynchronous updating with transient length > 1.

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14. The Role of the Hippocampus in Solving the Morris Water Maze

A. David Redish, David S. Touretzky
https://doi.org/10.1162/089976698300017908 | pp. 73-111

We suggest that the hippocampus plays two roles that allow rodents to solve the hidden-platform water maze: self-localization and route replay. When an animal explores an environment such as the water maze, the combination of place fields and correlational (Hebbian) long-term potentiation produces a weight matrix in the CA3 recurrent collaterals such that cells with overlapping place fields are more strongly interconnected than cells with nonoverlapping fields. When combined with global inhibition, this forms an attractor with coherent representations of position as stable states. When biased by local view information, this allows the animal to determine its position relative to the goal when it returns to the environment. We call this self-localization. When an animal traces specific routes within an environment, the weights in the CA3 recurrent collaterals become asymmetric. We show that this stores these routes in the recurrent collaterals. When primed with noise in the absence of sensory input, a coherent representation of position still forms in the CA3 population, but then that representation drifts, retracing a route. We show that these two mechanisms can coexist and form a basis for memory consolidation, explaining the anterograde and limited retrograde amnesia seen following hippocampal lesions.

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15. Near-Saddle-Node Bifurcation Behavior as Dynamics in Working Memory for Goal-Directed Behavior

Hiroyuki Nakahara, Kenji Doya
https://doi.org/10.1162/089976698300017917 | pp. 113-132

In consideration of working memory as a means for goal-directed behavior in nonstationary environments, we argue that the dynamics of working memory should satisfy two opposing demands: long-term maintenance and quick transition. These two characteristics are contradictory within the linear domain. We propose the near-saddle-node bifurcation behavior of a sigmoidal unit with a self-connection as a candidate of the dynamical mechanism that satisfies both of these demands. It is shown in evolutionary programming experiments that the near-saddle-node bifurcation behavior can be found in recurrent networks optimized for a task that requires efficient use of working memory. The results suggests that the near-saddle-node bifurcation behavior may be a functional necessity for survival in nonstationary environments.

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16. The Canonical Form of Nonlinear Discrete-Time Models

Gérard Dreyfus, Yizhak Idan
https://doi.org/10.1162/089976698300017926 | pp. 133-164

Discrete-time models of complex nonlinear processes, whether physical, biological, or economical, are usually under the form of systems of coupled difference equations. In analyzing such systems, one of the first tasks is to find a state-space description of the process—that is, a set of state variables and the associated state equations. We present a methodology for finding a set of state variables and a canonical representation of a class of systems described by a set of recurrent discrete-time, time-invariant equations. In the field of neural networks, this is of special importance since the application of standard training algorithms requires the network to be in a canonical form. Several illustrative examples are presented.

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17. A Low-Sensitivity Recurrent Neural Network

Andrew D. Back, Ah Chung Tsoi

https://doi.org/10.1162/089976698300017935 | pp. 165-188

The problem of high sensitivity in modeling is well known. Small perturbations in the model parameters may result in large, undesired changes in the model behavior. A number of authors have considered the issue of sensitivity in feedforward neural networks from a probabilistic perspective. Less attention has been given to such issues in recurrent neural networks. In this article, we present a new recurrent neural network architecture, that is capable of significantly improved parameter sensitivity properties compared to existing recurrent neural networks. The new recurrent neural network generalizes previous architectures by employing alternative discrete-time operators in place of the shift operator normally used. An analysis of the model demonstrates the existence of parameter sensitivity in recurrent neural networks and supports the proposed architecture. The new architecture performs significantly better than previous recurrent neural networks, as shown by a series of simple numerical experiments.

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18. Fixed-Point Attractor Analysis for a Class of Neurodynamics

Jianfeng Feng, David Brown

https://doi.org/10.1162/089976698300017944 | pp. 189-213

Nearly all models in neural networks start from the assumption that the input-output characteristic is a sigmoidal function. On parameter space, we present a systematic and feasible method for analyzing the whole spectrum of attractors—all-saturated, all-but-one-saturated, all-but-twosaturated, and so on—of a neurodynamical system with a saturated sigmoidal function as its input-output characteristic. We present an argument that claims, under a mild condition, that only all-saturated or all but-one-saturated attractors are observable for the neurodynamics. For any given all-saturated configuration [Formula: see text] (all-but-one-saturated configuration [Formula: see text]) the article shows how to construct an exact parameter region R([Formula: see text])([Formula: see text]([Formula: see text])) such that if and only if the parameters fall within R([Formula: see text])([Formula: see text]([Formula: see text])), then [Formula: see text]([Formula: see text]) is an attractor (a fixed point) of the dynamics. The parameter region for an all-saturated fixed-point attractor is independent of the specific choice of a saturated sigmoidal function, whereas for an all-but-one-saturated fixed point, it is sensitive to the input-output characteristic. Based on a similar idea, the role of weight normalization realized by a saturated sigmoidal function in competitive learning is discussed. A necessary and sufficient condition is provided to distinguish two kinds of competitive learning: stable competitive learning with the weight vectors representing extremes of input space and being fixed-point attractors, and unstable competitive learning. We apply our results to Linsker's model and (using extreme value theory in statistics) the Hopfield model and obtain some novel results on these two models.

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19. GTM: The Generative Topographic Mapping

Chris Bishop, Markus Svensén, Christopher K. I. Williams

https://doi.org/10.1162/089976698300017953 | pp. 215-234

Latent variable models represent the probability density of data in a space of several dimensions in terms of a smaller number of latent, or hidden, variables. A familiar example is factor analysis, which is based on a

linear transformation between the latent space and the data space. In this article, we introduce a form of nonlinear latent variable model called the generative topographic mapping, for which the parameters of the model can be determined using the expectation-maximization algorithm. GTM provides a principled alternative to the widely used self-organizing map (SOM) of Kohonen (1982) and overcomes most of the significant limitations of the SOM. We demonstrate the performance of the GTM algorithm on a toy problem and on simulated data from flow diagnostics for a multiphase oil pipeline.

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20. Identification Criteria and Lower Bounds for Perceptron-Like Learning Rules

Michael Schmitt

https://doi.org/10.1162/089976698300017962 | pp. 235-250

Perceptron-like learning rules are known to require exponentially many correction steps in order to identify boolean threshold functions exactly. We introduce criteria that are weaker than exact identification and investigate whether learning becomes significantly faster if exact identification is replaced by one of these criteria: probably approximately correct (PAC) identification, order identification, and sign identification. PAC identification is based on the learning paradigm introduced by Valiant (1984) and is known to be easier than exact identification. Order identification uses the fact that each threshold function induces an ordering relation on the input variables that can be represented by weights of linear size. Sign identification is based on a property of threshold functions known as unateness and requires only weights of constant size. We show that Perceptron-like learning rules cannot satisfy these criteria when the number of correction steps is to be bounded by a polynomial. We also present an exponential lower bound for order identification with the learning rules introduced by Littlestone (1988). Our results show that efficiency imposes severe restrictions on what can be learned with local learning rules.

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21. Natural Gradient Works Efficiently in Learning

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https://doi.org/10.1162/089976698300017746 | pp. 251-276

When a parameter space has a certain underlying structure, the ordinary gradient of a function does not represent its steepest direction, but the natural gradient does. Information geometry is used for calculating the natural gradients in the parameter space of perceptrons, the space of matrices (for blind source separation), and the space of linear dynamical systems (for blind source deconvolution). The dynamical behavior of natural gradient online learning is analyzed and is proved to be Fisher efficient, implying that it has asymptotically the same performance as the optimal batch estimation of parameters. This suggests that the plateau phenomenon, which appears in the backpropagation learning algorithm of multilayer perceptrons, might disappear or might not be so serious when the natural gradient is used. An adaptive method of updating the learning rate is proposed and analyzed.

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22. Adding Lateral Inhibition to a Simple Feedforward Network Enables It to Perform Exclusive-Or

Leslie S. Smith
https://doi.org/10.1162/089976698300017755 | pp. 277-280

A simple laterally inhibited recurrent network that implementse xclusive-or is demonstrated. The network consists of two mutually inhibitory units with logistic output function, each receiving one external input and each connected to a simple threshold output unit. The mutually inhibitory units settle into a point attractor. We investigate the range of steepness of the logistic and the range of inhibitory weights for which the network can perform exclusive-or.

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23. Combined Learning and Use for a Mixture Model Equivalent to the RBF Classifier

David J. Miller, Hasan S. Uyar
https://doi.org/10.1162/089976698300017764 | pp. 281-293

We show that the decision function of a radial basis function (RBF) classifier is equivalent in form to the Bayes-optimal discriminant associated with a special kind of mixture-based statistical model. The relevant mixture model is a type of mixture-of-experts model for which class labels, like continuous-valued features, are assumed to have been generated randomly, conditional on the mixture component of origin. The new interpretation shows that RBF classifiers effectively assume a probability model, which, moreover, is easily determined given the designed RBF. This interpretation also suggests a statistical learning objective as an alternative to standard methods for designing the RBF-equivalent models. The statistical objective is especially useful for incorporating unlabeled data to enhance learning. Finally, it is observed that any new data to classify are simply additional unlabeled data. Thus, we suggest a combined learning and use paradigm, to be invoked whenever there are new data to classify.

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24. Modeling the Surround of MT Cells and Their Selectivity for Surface Orientation in Depth Specified by Motion

Lin Liu, Marc M. Van Hulle
https://doi.org/10.1162/089976698300017773 | pp. 295-312

The projective transformation onto the retina loses the explicit 3D shape description of a moving object. Theoretical studies show that the reconstruction of 3D shape from 2D motion information (shape from motion, SFM) is feasible provided that the first- and second-order directional derivatives of the 2D velocity field are available. Experimental recordings have revealed that the receptive fields of the majority of the cells in macaque area middle temporal (MT) display an antagonistic (suppressive) surround and that a sizable portion of these surrounds are asymmetrical. This has led to the conjecture that these cells provide a local measure for the directional derivatives of the 2D velocity field. In this article, we adopt a nonparametric and biologically plausible approach to modeling the role played by the MT surrounds in the recovery of the orientation in depth (the slant and tilt) of a moving (translating) plane. A three-layered neural network is trained to represent the slant and tilt from the projected motion vectors. The hidden units of the network have speed-tuning characteristics and represent the MT model neurons with their surrounds. We conjecture that the MT surround results from lateral inhibitory connections with other MT cells and that populations of these cells, with different surround types, code linearly for slant and tilt of translating planes.

25. A Self-Organizing Neural Network Architecture for Navigation Using Optic Flow

Seth Cameron, Stephen Grossberg, Frank H. Guenther

https://doi.org/10.1162/089976698300017782 | pp. 313-352

This article describes a self-organizing neural network architecture that transforms optic flow and eye position information into representations of heading, scene depth, and moving object locations. These representations are used to navigate reactively in simulations involving obstacle avoidance and pursuit of a moving target. The network's weights are trained during an action-perception cycle in which self-generated eye and body movements produce optic flow information, thus allowing the network to tune itself without requiring explicit knowledge of sensor geometry. The confounding effect of eye movement during translation is suppressed by learning the relationship between eye movement outflow commands and the optic flow signals that they induce. The remaining optic flow field is due to only observer translation and independent motion of objects in the scene. A self-organizing feature map categorizes normalized translational flow patterns, thereby creating a map of cells that code heading directions. Heading information is then recombined with translational flow patterns in two different ways to form maps of scene depth and moving object locations. Most of the learning processes take place concurrently and evolve through unsupervised learning. Mapping the learned heading representations onto heading labels or motor commands requires additional structure. Simulations of the network verify its performance using both noise-free and noisy optic flow information.

26. Analysis of Direction Selectivity Arising from Recurrent Cortical Interactions

Paul Mineiro, David Zipser

https://doi.org/10.1162/089976698300017791 | pp. 353-371

The relative contributions of feedforward and recurrent connectivity to the direction-selective responses of cells in layer IVB of primary visual cortex are currently the subject of debate in the neuroscience community. Recently, biophysically detailed simulations have shown that realistic direction-selective responses can be achieved via recurrent cortical interactions between cells with nondirection-selective feedforward input (Suarez et al., 1995; Maex & Orban, 1996). Unfortunately these models, while desirable for detailed comparison with biology, are complex and thus difficult to analyze mathematically. In this article, a relatively simple cortical dynamical model is used to analyze the emergence of direction-selective responses via recurrent interactions. A comparison between a model based on our analysis and physiological data is presented. The approach also allows analysis of the recurrently propagated signal, revealing the predictive nature of the implementation.

27. Statistically Efficient Estimation Using Population Coding

Alexandre Pouget, Kechen Zhang, Sophie Denève, Peter E. Latham

https://doi.org/10.1162/089976698300017809 | pp. 373-401

Coarse codes are widely used throughout the brain to encode sensory and motor variables. Methods designed to interpret these codes, such as population vector analysis, are either inefficient (the variance of the estimate is much larger than the smallest possible variance) or biologically implausible, like maximum likelihood. Moreover, these methods attempt to compute a scalar or vector estimate of the encoded variable. Neurons are faced with a similar estimation problem. They must read out the responses of the presynaptic neurons, but, by contrast, they typically encode the variable with a further population code rather than as a scalar. We show how a nonlinear recurrent network can be used to perform estimation in a near-optimal way while keeping the estimate in a coarse code format. This work suggests that lateral connections in the cortex may be involved in cleaning up uncorrelated noise among neurons representing similar variables.

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28. Probabilistic Interpretation of Population Codes

Richard S. Zemel, Peter Dayan, Alexandre Pouget

https://doi.org/10.1162/089976698300017818 | pp. 403-430

We present a general encoding-decoding framework for interpreting the activity of a population of units. A standard population code interpretation method, the Poisson model, starts from a description as to how a single value of an underlying quantity can generate the activities of each unit in the population. In casting it in the encoding-decoding framework, we find that this model is too restrictive to describe fully the activities of units in population codes in higher processing areas, such as the medial temporal area. Under a more powerful model, the population activity can convey information not only about a single value of some quantity but also about its whole distribution, including its variance, and perhaps even the certainty the system has in the actual presence in the world of the entity generating this quantity. We propose a novel method for forming such probabilistic interpretations of population codes and compare it to the existing method.

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29. Stable and Rapid Recurrent Processing in Realistic Autoassociative Memories

Francesco P. Battaglia, Alessandro Treves

https://doi.org/10.1162/089976698300017827 | pp. 431-450

It is shown that in those autoassociative memories that learn by storing multiple patterns of activity on their recurrent collateral connections, there is a fundamental conflict between dynamical stability and storage capacity. It is then found that the network can nevertheless retrieve many different memory patterns, as predicted by nondynamical analyses, if its firing is regulated by inhibition that is sufficiently multiplicative in nature. Simulations of a model network with integrate-and-fire units confirm that this is a realistic solution to the conflict. The simulations also confirm the earlier analytical result that cued-elicited memory retrieval, which follows an exponential time course, occurs in a time linearly related to the time constant for synaptic conductance inactivation and relatively independent of neuronal time constants and firing levels.

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30. Synaptic Runaway in Associative Networks and the Pathogenesis of Schizophrenia

Asnat Greenstein■Messica, Eytan Ruppin
https://doi.org/10.1162/089976698300017836 | pp. 451-465

Synaptic runaway denotes the formation of erroneous synapses and premature functional decline accompanying activity-dependent learning in neural networks. This work studies synaptic runaway both analytically and numerically in binary-firing associative memory networks. It turns out that synaptic runaway is of fairly moderate magnitude in these networks under normal, baseline conditions. However, it may become extensive if the threshold for Hebbian learning is reduced. These findings are combined with recent evidence for arrested N-methyl-D-aspartate (NMDA) maturation in schizophrenics, to formulate a new hypothesis concerning the pathogenesis of schizophrenic psychotic symptoms in neural terms.

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31. On Numerical Simulations of Integrate-and-Fire Neural Networks

D. Hansel, Germán Mato, C. Meunier, L. Neltner
https://doi.org/10.1162/089976698300017845 | pp. 467-483

It is shown that very small time steps are required to reproduce correctly the synchronization properties of large networks of integrate-and-fire neurons when the differential system describing their dynamics is integrated with the standard Euler or second-order Runge-Kutta algorithms. The reason for that behavior is analyzed, and a simple improvement of these algorithms is proposed.

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32. A Floating-Gate MOS Implementation of Resistive Fuse

T. Matsumoto, Toshiaki Sawaji, Toshikatsu Sakai, Hiromasa Nagai
https://doi.org/10.1162/089976698300017854 | pp. 485-498

Resistive fuses are key elements in weak string filters, which smooth out noise while detecting step edges inherent in original data. A resistive fuse is implemented by two pairs of floating-gate metal oxide semiconductor (MOS) transistors in a chip by a standard double poly complementary MOS process.

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33. Toward a Biophysically Plausible Bidirectional Hebbian Rule

Norberto M. Grzywacz, Pierre■Yves Burgi
https://doi.org/10.1162/089976698300017629 | pp. 499-520

Although the commonly used quadratic Hebbian-anti-Hebbian rules lead to successful models of plasticity and learning, they are inconsistent with neurophysiology. Other rules, more physiologically plausible, fail to specify the biological mechanism of bidirectionality and the biological mechanism that prevents synapses from changing from excitatory to inhibitory, and vice versa. We developed a synaptic bidirectional Hebbian rule that does not suffer from these problems. This rule was compared with physiological homosynaptic conditions in the hippocampus, with the results indicating the consistency of this rule with long-term potentiation (LTP) and long-term depression (LTD) phenomenologies. The phenomenologies considered included the reversible dynamics of LTP and LTD and the effects of N-methyl-D-aspartate blockers and phosphatase inhibitors.

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34. Axon Guidance: Stretching Gradients to the Limit

Geoffrey J. Goodhill, Herwig Baier

https://doi.org/10.1162/089976698300017638 | pp. 521-527

Neuronal growth cones, the sensory-motile structures at the tips of developing axons, navigate to their targets over distances that can be many times greater than their diameter. They may accomplish this impressive task by following spatial gradients of axon guidance molecules in their environment (Bonhoeffer & Gierer, 1984; Tessier-Lavigne & Placzek, 1991; Baier & Bonhoeffer, 1994). We calculate the optimal shape of a gradient and the distance over which it can be detected by a growth cone for two competing mechanistic models of axon guidance. The results are surprisingly simple: Regardless of the mechanism, the maximum distance is about 1 cm. Since gradients and growth cones have coevolved, we suggest that the shape of the gradient in situ will predict the mechanism of gradient detection. In addition, we show that the experimentally determined dissociation constants for receptor-ligand complexes implicated in axon guidance are about optimal with respect to maximizing guidance distance. The relevance of these results to the retinotectal system is discussed.

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35. Equivalence of a Sprouting-and-Retraction Model and Correlation-Based Plasticity Models of Neural Development

Kenneth D. Miller

https://doi.org/10.1162/089976698300017647 | pp. 529-547

A simple model of correlation-based synaptic plasticity via axonal sprouting and retraction (Elliott, Howarth, & Shadbolt, 1996a) is shown to be equivalent to the class of correlation-based models (Miller, Keller, & Stryker, 1989), although these were formulated in terms of weight modification of anatomically fixed synapses. Both models maximize the same measure of synaptic correlation, subject to certain constraints on connectivity. Thus, the analyses of the correlation-based models suffice to characterize the behavior of the sprouting-and-retraction model. More detailed models are needed for theoretical distinctions to be drawn between plasticity via sprouting and retraction, weight modification, or a combination. The model of Elliott et al. involves stochastic search through allowed weight patterns for those that improve correlations. That of Miller et al. instead follows dynamical equations that determine continuous changes of the weights that improve correlations. The identity of these two approaches is shown to depend on the use of subtractive constraint enforcement in the models of Miller et al. More generally, to model the idea that neural development acts to maximize some measure of correlation subject to a constraint on the summed synaptic weight, the constraint must be enforced subtractively in a dynamical model.

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36. Axonal Processes and Neural Plasticity: A Reply

Terry Elliott, Clare Howarth, N. R. Shadbolt

https://doi.org/10.1162/089976698300017656 | pp. 549-554

We examine the claim that a class of sprouting-and-retraction models is mathematically equivalent to a fixed-anatomy model. We accept, subject to important caveats, a narrow mathematical equivalence of the energy functions in both classes of model. We argue that this narrow equivalence of energy functions does not, however, entail equivalence of the models. Indeed, the claim of complete model equivalence hides

significant dynamical differences between the approaches, which we discuss. We also disagree that our work demonstrates that subtractive constraint enforcement is natural in fixed-anatomy models.

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37. Synaptic Delay Learning in Pulse-Coupled Neurons

Harald Hüning, Helmut Glünder, Günther Palm

https://doi.org/10.1162/089976698300017665 | pp. 555-565

We present rules for the unsupervised learning of coincidence between excitatory postsynaptic potentials (EPSPs) by the adjustment of postsynaptic delays between the transmitter binding and the opening of ion channels. Starting from a gradient descent scheme, we develop a robust and more biological threshold rule by which EPSPs from different synapses can be gradually pulled into coincidence. The synaptic delay changes are determined from the summed potential--at the site where the coincidence is to be established--and from postulated synaptic learning functions that accompany the individual EPSPs. According to our scheme, templates for the detection of spatiotemporal patterns of synaptic activation can be learned, which is demonstrated by computer simulation. Finally, we discuss possible relations to biological mechanisms.

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38. Neural Processing in the Subsecond Time Range in the Temporal Cortex

Kiyohiko Nakamura

https://doi.org/10.1162/089976698300017674 | pp. 567-595

The hypothesis that cortical processing of the millisecond time range is performed by latency competition between the first spikes produced by neuronal populations is analyzed. First, theorems that describe how the mechanism of latency competition works in a model cortex are presented. The model is a sequence of cortical areas, each of which is an array of neuronal populations that laterally inhibit each other. Model neurons are integrate-and-fire neurons. Second, the model is applied to the ventral pathway of the temporal lobe, and neuronal activity of the superior temporal sulcus of the monkey is reproduced with the model pathway. It consists of seven areas: V1, V2/V3, V4, PIT, CIT, AIT, and STPa. Neural activity predicted with the model is compared with empirical data. There are four main results: (1) Neural responses of the area STPa of the model showed the same fast discrimination between stimuli that the corresponding responses of the monkey did: both were significant within 5 ms of the response onset. (2) The hypothesis requires that the response latency of cortical neurons should be shorter for stronger responses. This requirement was verified by both the model simulation and the empirical data. (3) The model reproduced fast discrimination even when spontaneous random firing of 9 Hz was introduced to all the cells. This suggests that the latency competition performed by neuronal populations is robust. (4) After the first few competitions, the mechanism of latency competition always detected the strongest of input activations with different latencies.

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39. Temporal-Code to Rate-Code Conversion by Neuronal Phase-Locked Loops

Ehud Ahissar

https://doi.org/10.1162/089976698300017683 | pp. 597-650

Peripheral sensory activity follows the temporal structure of input signals. Central sensory processing uses also rate coding, and motor outputs appear to be primarily encoded by rate. I propose here a simple, efficient structure, converting temporal coding to rate coding by neuronal phase-locked loops (PLL). The simplest form of a PLL includes a phase detector (that is, a neuronal-plausible version of an ideal coincidence detector) and a controllable local oscillator that are connected in a negative feedback loop. The phase detector compares the firing times of the local oscillator and the input and provides an output whose firing rate is monotonically related to the time difference. The output rate is fed back to the local oscillator and forces it to phase-lock to the input. Every temporal interval at the input is associated with a specific pair of output rate and time difference values; the higher the output rate, the further the local oscillator is driven from its intrinsic frequency. Sequences of input intervals, which by definition encode input information, are thus represented by sequences of firing rates at the PLL's output. The most plausible implementation of PLL circuits is by thalamocortical loops in which populations of thalamic "relay" neurons function as phase detectors that compare the timings of cortical oscillators and sensory signals. The output in this case is encoded by the thalamic population rate. This article presents and analyzes the algorithmic and the implementation levels of the proposed PLL model and describes the implementation of the PLL model to the primate tactile system.

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40. Deformation Theory of Dynamic Link Matching

Toru Aonishi, Koji Kurata

https://doi.org/10.1162/089976698300017692 | pp. 651-669

Dynamic link matching is a self&hyphenorganizing topographic mapping between a template image and a data image. The mapping tends to be continuous, linking two points sharing similar local features, which, as a result, can lead to its deformation to some degree. In analyzing such deformation mathematically, we reduced the model equation to a phase equation, which enabled us to clarify the principles of the deformation process and the relationship between high&hyphendimensional models and low&hyphendimensional ones. We also elucidated the characteristics of the model in the context of the standard regularization theory.

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41. Constrained Optimization for Neural Map Formation: A Unifying Framework for Weight Growth and Normalization

Laurenz Wiskott, Terrence J. Sejnowski

https://doi.org/10.1162/089976698300017700 | pp. 671-716

Computational models of neural map formation can be considered on at least three different levels of abstraction: detailed models including neural activity dynamics, weight dynamics that abstract from the neural activity dynamics by an adiabatic approximation, and constrained optimization from which equations governing weight dynamics can be derived. Constrained optimization uses an objective function, from which a weight growth rule can be derived as a gradient flow, and some constraints, from which normalization rules are derived. In this article, we present an example of how an optimization problem can be derived from detailed nonlinear neural dynamics. A systematic investigation reveals how different weight dynamics introduced previously can be derived from two types of objective function terms and two types of constraints. This includes dynamic link matching as a special case of neural map formation. We focus in particular on the

role of coordinate transformations to derive different weight dynamics from the same optimization problem. Several examples illustrate how the constrained optimization framework can help in understanding, generating, and comparing different models of neural map formation. The techniques used in this analysis may also be useful in investigating other types of neural dynamics.

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42. Breaking Rotational Symmetry in a Self-Organizing Map Model for Orientation Map Development

Maximilian Riesenhuber, H.-U. Bauer, D. Brockmann, T. Geisel

https://doi.org/10.1162/089976698300017719 | pp. 717-730

We analyze the pattern formation behavior of a high&hyphendimensional self&hyphenorganizing map (SOM) model for the competitive projection of ON&hyphencenter&hyphentype and OFF&hyphencenter&hyphentype inputs to a common map layer. We mathematically show, and numerically confirm, that even isotropic stimuli can drive the development of oriented receptive fields and an orientation map in this model. This result provides an important missing link in the spectrum of pattern formation behaviors observed in SOM models. Extending the model by including further layers for binocular inputs, we also investigate the combined development of orientation and ocular dominance maps. A parameter region for combined patterns exists; corresponding maps show a preference for perpendicular intersection angles between iso&hyphenorientation lines and ocularity domain boundaries, consistent with experimental observations.

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43. Nonlinear Time-Series Prediction with Missing and Noisy Data

Volker Tresp, Reimar Hofmann

https://doi.org/10.1162/089976698300017728 | pp. 731-747

We derive solutions for the problem of missing and noisy data in nonlinear time&hyphenseries prediction from a probabilistic point of view. We discuss different approximations to the solutions &hyphen in particular, approximations that require either stochastic simulation or the substitution of a single estimate for the missing data. We show experimentally that commonly used heuristics can lead to suboptimal solutions. We show how error bars for the predictions can be derived and how our results can be applied to K&hyphenstep prediction. We verify our solutions using two chaotic time series and the sunspot data set. In particular, we show that for K&hyphenstep prediction, stochastic simulation is superior to simply iterating the predictor.

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44. Issues in Bayesian Analysis of Neural Network Models

Peter Müller, David R■■os Insua

https://doi.org/10.1162/089976698300017737 | pp. 749-770

Stemming from work by Buntine and Weigend (1991) and MacKay (1992), there is a growing interest in Bayesian analysis of neural network models. Although conceptually simple, this problem is computationally involved. We suggest a very efficient Markov chain Monte Carlo scheme for inference and prediction with fixed&hyphenarchitecture feedforward neural networks. The scheme is then extended to the variable architecture case, providing a data&hyphendriven procedure to identify sensible architectures.

45. Computational Models of Neuromodulation

Jean■Marc Fellous, Christiane Linster

https://doi.org/10.1162/089976698300017476 | pp. 771-805

Computational modeling of neural substrates provides an excellent theoretical framework for the understanding of the computational roles of neuromodulation. In this review, we illustrate, with a large number of modeling studies, the specific computations performed by neuromodulation in the context of various neural models of invertebrate and vertebrate preparations. We base our characterization of neuromodulations on their computational and functional roles rather than on anatomical or chemical criteria. We review the main framework in which neuromodulation has been studied theoretically (central pattern generation and oscillations, sensory processing, memory and information integration). Finally, we present a detailed mathematical overview of how neuromodulation has been implemented at the single cell and network levels in modeling studies. Overall, neuromodulation is found to increase and control computational complexity.

46. Weight-Value Convergence of the SOM Algorithm for Discrete Input

Siming Lin, Jennie Si

https://doi.org/10.1162/089976698300017485 | pp. 807-814

Some insights on the convergence of the weight values of the self-organizing map (SOM) to a stationary state in the case of discrete input are provided. The convergence result is obtained by applying the Robbins-Monro algorithm and is applicable to input-output maps of any dimension.

47. Reading Neuronal Synchrony with Depressing Synapses

Walter Senn, Idan Segev, Misha Tsodyks

https://doi.org/10.1162/089976698300017494 | pp. 815-819

A recent experiment showed that neurons in the primary auditory cortex of the monkey do not change their mean firing rate during an ongoing tone stimulus. The only change was an enhanced correlation among the individual spike trains during the tone. We show that there is an easy way to extract this coherence information in the cortical cell population by projecting the spike trains through depressing synapses onto a postsynaptic neuron.

48. Neural Networks with Dynamic Synapses

Misha Tsodyks, Klaus Pawelzik, Henry Markram

https://doi.org/10.1162/089976698300017502 | pp. 821-835

Transmission across neocortical synapses depends on the frequency of presynaptic activity (Thomson & Deuchars, 1994). Interpyramidal synapses in layer V exhibit fast depression of synaptic transmission, while other types of synapses exhibit facilitation of transmission. To study the role of dynamic synapses in network computation, we propose a unified phenomenological model that allows computation of the postsynaptic

current generated by both types of synapses when driven by an arbitrary pattern of action potential (AP) activity in a presynaptic population. Using this formalism, we analyze different regimes of synaptic transmission and demonstrate that dynamic synapses transmit different aspects of the presynaptic activity depending on the average presynaptic frequency. The model also allows for derivation of mean-field equations, which govern the activity of large, interconnected networks. We show that the dynamics of synaptic transmission results in complex sets of regular and irregular regimes of network activity.

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49. Spike Frequency Adaptation Affects the Synchronization Properties of Networks of Cortical Oscillators

Sharon Crook, Bard Ermentrout, James M. Bower

https://doi.org/10.1162/089976698300017511 | pp. 837-854

Oscillations in many regions of the cortex have common temporal characteristics with dominant frequencies centered around the 40 Hz (gamma) frequency range and the 5-10 Hz (theta) frequency range. Experimental results also reveal spatially synchronous oscillations, which are stimulus dependent (Gray & Singer, 1987; Gray, Konig, Engel, & Singer, 1989; Engel, Konig, Kreiter, Schillern, & Singer, 1992). This rhythmic activity suggests that the coherence of neural populations is a crucial feature of cortical dynamics (Gray, 1994). Using both simulations and a theoretical coupled oscillator approach, we demonstrate that the spike frequency adaptation seen in many pyramidal cells plays a subtle but important role in the dynamics of cortical networks. Without adaptation, excitatory connections among model pyramidal cells are desynchronizing. However, the slow processes associated with adaptation encourage stable synchronous behavior.

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50. GABAergic Inhibitory Control of the Transient and Sustained Components of Orientation Selectivity in a Model Microcolumn in Layer 4 of Cat Visual Cortex

Paul C. Bush, Nicholas J. Priebe

https://doi.org/10.1162/089976698300017520 | pp. 855-867

Recently proposed models of orientation tuning in layer 4 of cat primary visual cortex (Somers, Nelson, & Sur, 1995; Douglas, Koch, Mahowald, Martin, & Suarez, 1995) rely on widespread inhibitory intracortical connections to suppress the nonoptimal component of a broadly tuned thalamic input, while local excitatory intracortical connections amplify the optimal component. However, new experimental data (Ferster, Chung, & Wheat, 1996) and theoretical analyses (Ferster, 1987; Krukowski, Priebe, & Miller, 1996) show that the temporally modulated component of thalamic input is well tuned and that the cortical circuitry must simply subtract an unmodulated DC component at nonoptimal orientations to obtain sharp tuning. In addition, within a single hypercolumn in layer 4, inhibitory and excitatory layer 4 neurons have approximately equal-sized axonal fields, making the most of their synapses within their own dendritic field (Kisvarday, Martin, Whitteridge, & Somogyi, 1985; Martin & Whitteridge, 1984). We have constructed a model of a single microcolumn in which GABA inhibition subtracts the DC and controls the sustained response, while GABA inhibition controls the response to transient and suprathreshold inputs. The model fits experimental data based on stimulation with drifting sine-wave gratings as well as flashed bars, explains a counterintuitive property of the GABA conductance, and at suboptimal orientations and submaximal contrasts produces an

exponential distribution of firing frequencies.

51. Changes in GABA_BModulation During a Theta Cycle May Be Analogous to the Fall of Temperature During Annealing

Vikaas S. Sohal, Michael E. Hasselmo

https://doi.org/10.1162/089976698300017539 | pp. 869-882

Changes in GABA modulation may underlie experimentally observed changes in the strength of synaptic transmission at different phases of the theta rhythm (Wyble, Linster, & Hasselmo, 1997). Analysis demonstrates that these changes improve sequence disambiguation by a neural network model of CA3. We show that in the framework of Hopfield and Tank (1985), changes in GABA suppression correspond to changes in the effective temperature and the relative energy of data terms and constraints of an analog network. These results suggest that phasic changes in the activity of inhibitory interneurons during a theta cycle may produce dynamics that resemble annealing. These dynamics may underlie a role for the theta cycle in improving sequence retrieval for spatial navigation.

52. RST: A Connectionist Architecture to Deal with Spatiotemporal Relationships

Jean-Cédric Chappelier, Alain Grumbach

https://doi.org/10.1162/089976698300017548 | pp. 883-902

In the past decade, connectionism has proved its efficiency in the field of static pattern recognition. The next challenge is to deal with spatiotemporal problems. This article presents a new connectionist architecture, RST (réseau spatio temporel [spatio temporal network]), with such spatiotemporal capacities. It aims at taking into account at the architecture level both spatial relationships (e.g., as between neighboring pixels in an image) and temporal relationships (e.g., as between consecutive images in a video sequence). Concerning the spatial aspect, the network is embedded in actual space (two) or three-dimensional-, the metrics of which directly influence its structure through a connection distribution function. For the temporal aspect, we looked toward biology and used a leaky-integrator neuron model with a refractory period and postsynaptic potentials. The propagation of activity by spatiotemporal synchronized waves enables RST to perform motion detection and localization in sequences of video images.

53. A Neural Model of Contour Integration in the Primary Visual Cortex

Li Zhaoping

https://doi.org/10.1162/089976698300017557 | pp. 903-940

Experimental observations suggest that contour integration may take place in V1. However, there has yet to be a model of contour integration that uses only known V1 elements, operations, and connection patterns. This article introduces such a model, using orientation selective cells, local cortical circuits, and horizontal intracortical connections. The model is composed of recurrently connected excitatory neurons and inhibitory interneurons, receiving visual input via oriented receptive fields resembling those found in primary visual cortex. Intracortical interactions modify initial activity patterns from input, selectively amplifying the activities

of edges that form smooth contours in the image. The neural activities produced by such interactions are oscillatory and edge segments within a contour oscillate in synchrony. It is shown analytically and empirically that the extent of contour enhancement and neural synchrony increases with the smoothness, length, and closure of contours, as observed in experiments on some of these phenomena. In addition, the model incorporates a feedback mechanism that allows higher visual centers selectively to enhance or suppress sensitivities to given contours, effectively segmenting one from another. The model makes the testable prediction that the horizontal cortical connections are more likely to target excitatory (or inhibitory) cells when the two linked cells have their preferred orientation aligned with (or orthogonal to) their relative receptive field center displacements.

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54. Predictive Neural Networks for Learning the Time Course of Blood Glucose Levels from the Complex Interaction of Counterregulatory Hormones

Klaus Prank, Clemens Jürgens, A. von zur Mühlen, Georg Brabant

https://doi.org/10.1162/089976698300017566 | pp. 941-953

Diabetes mellitus is a widespread disease associated with an impaired hormonal regulation of normal blood glucose levels. Patients with insulin-dependent diabetes mellitus (IDDM) who practice conventional insulin therapy are at risk of developing hypoglycemia (low levels of blood glucose), which can lead to severe dysfunction of the central nervous system. In large retrospective studies, up to approximately 4% of deaths of patients with IDDM have been attributed to hypoglycemia (Cryer, Fisher, & Shamoon, 1994; Tunbridge, 1981; Deckert, Poulson, & Larsen, 1978). Thus, a better understanding of the complex hormonal interaction preventing hypoglycemia is crucial for treatment. Experimental data from a study on insulin-induced hypoglycemia in healthy subjects are used to demonstrate that feedforward neural networks are capable of predicting the time course of blood glucose levels from the complex interaction of glucose counterregulatory (glucose-raising) hormones and insulin. By simulating the deficiency of single hormonal factors in this regulatory network, we found that the predictive impact of glucagon, epinephrine, and growth hormone secretion, but not of cortisol and norepinephrine, were dominant in restoring normal levels of blood glucose following hypoglycemia.

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55. Properties of Support Vector Machines

Massimiliano Pontil, Alessandro Verri

https://doi.org/10.1162/089976698300017575 | pp. 955-974

Support vector machines (SVMs) perform pattern recognition between two point classes by finding a decision surface determined by certain points of the training set, termed support vectors (SV). This surface, which in some feature space of possibly infinite dimension can be regarded as a hyperplane, is obtained from the solution of a problem of quadratic programming that depends on a regularization parameter. In this article, we study some mathematical properties of support vectors and show that the decision surface can be written as the sum of two orthogonal terms, the first depending on only the margin vectors (which are SVs lying on the margin), the second proportional to the regularization parameter. For almost all values of the parameter, this enables us to predict how the decision surface varies for small parameter changes. In the special but important case of feature space of finite dimension m, we also show that m + 1 SVs are usually

sufficient to determine the decision surface fully. For relatively small m, this latter result leads to a consistent reduction of the SV number.

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56. Validation of Voting Committees

Eric Bax

https://doi.org/10.1162/089976698300017584 | pp. 975-986

This article contains a method to bound the test errors of voting committees with members chosen from a pool of trained classifiers. There are so many prospective committees that validating them directly does not achieve useful error bounds. Because there are fewer classifiers than prospective committees, it is better to validate the classifiers individually than use linear programming to infer committee error bounds. We test the method using credit card data. Also, we extend the method to infer bounds for classifiers in general.

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57. Toward Optimally Distributed Computation

Peter Edwards, Alan F. Murray

https://doi.org/10.1162/089976698300017593 | pp. 987-1005

This article introduces the concept of optimally distributed computation in feedforward neural networks via regularization of weight saliency. By constraining the relative importance of the parameters, computation can be distributed thinly and evenly throughout the network. We propose that this will have beneficial effects on fault-tolerance performance and generalization ability in large network architectures. These theoretical predictions are verified by simulation experiments on two problems: one artificial and the other a real-world task. In summary, this article presents regularization terms for distributing neural computation optimally.

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58. Efficient Adaptive Learning for Classification Tasks with Binary Units

J. Manuel Torres Moreno, Mirta B. Gordon

https://doi.org/10.1162/089976698300017601 | pp. 1007-1030

This article presents a new incremental learning algorithm for classification tasks, called NetLines, which is well adapted for both binary and real-valued input patterns. It generates small, compact feedforward neural networks with one hidden layer of binary units and binary output units. A convergence theorem ensures that solutions with a finite number of hidden units exist for both binary and real-valued input patterns. An implementation for problems with more than two classes, valid for any binary classifier, is proposed. The generalization error and the size of the resulting networks are compared to the best published results on well-known classification benchmarks. Early stopping is shown to decrease overfitting, without improving the generalization performance.

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59. A Systematic and Effective Supervised Learning Mechanism Based on Jacobian Rank Deficiency

Guian Zhou, Jennie Si

https://doi.org/10.1162/089976698300017610 | pp. 1031-1045

Most neural network applications rely on the fundamental approximation property of feedforward networks. Supervised learning is a means of implementing this approximate mapping. In a realistic problem setting, a mechanism is needed to devise this learning process based on available data, which encompasses choosing an appropriate set of parameters in order to avoid overfitting, using an efficient learning algorithm measured by computation and memory complexities, ensuring the accuracy of the training procedures as measured by the training error, and testing and cross-validation for generalization. We develop a comprehensive supervised learning algorithm to address these issues. The algorithm combines training and pruning into one procedure by utilizing a common observation of Jacobian rank deficiency in feedforward networks. The algorithm not only reduces the training time and overall complexity but also achieves training accuracy and generalization capabilities comparable to more standard approaches. Extensive simulation results are provided to demonstrate the effectiveness of the algorithm.

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60. Dynamics of Membrane Excitability Determine Interspike Interval Variability: A Link Between Spike Generation Mechanisms and Cortical Spike Train Statistics

Boris Gutkin, Bard Ermentrout

https://doi.org/10.1162/089976698300017331 | pp. 1047-1065

We propose a biophysical mechanism for the high interspike interval variability observed in cortical spike trains. The key lies in the nonlinear dynamics of cortical spike generation, which are consistent with type I membranes where saddle-node dynamics underlie excitability (Rinzel & Ermentrout, 1989). We present a canonical model for type I membranes, the theta-neuron. The theta-neuron is a phase model whose dynamics reflect salient features of type I membranes. This model generates spike trains with coefficient of variation (CV) above 0.6 when brought to firing by noisy inputs. This happens because the timing of spikes for a type I excitable cell is exquisitely sensitive to the amplitude of the suprathreshold stimulus pulses. A noisy input current, giving random amplitude "kicks" to the cell, evokes highly irregular firing across a wide range of firing rates; an intrinsically oscillating cell gives regular spike trains. We corroborate the results with simulations of the Morris-Lecar (M-L) neural model with random synaptic inputs: type I M-L yields high CVs. When this model is modified to have type II dynamics (periodicity arises via a Hopf bifurcation), however, it gives regular spike trains (CV below 0.3). Our results suggest that the high CV values such as those observed in cortical spike trains are an intrinsic characteristic of type I membranes driven to firing by "random" inputs. In contrast, neural oscillators or neurons exhibiting type II excitability should produce regular spike trains.

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61. Correction to Proof That Recurrent Neural Networks Can Robustly Recognize Only Regular Languages

M. Casey

https://doi.org/10.1162/089976698300017340 | pp. 1067-1069

Our earlier article, “The Dynamics of Discrete-Time Computation, with Application to Recurrent Neural Networks and Finite State Machine Extraction” (Casey, 1996), contains a corollary that shows that finite

dimensional recurrent neural networks with noise in their state variables that perform algorithmic computations can perform only finite state machine computations. The proof of the corollary is technically incorrect. The problem resulted from the fact that the proof of the theorem on which the corollary is based was more general than the statement of the theorem, and it was the contents of the proof rather than the statement that were used to prove the corollary. In this note, we state the theorem in the necessary generality and then give the corrected proof of the corollary.

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62. On the Effect of Analog Noise in Discrete-Time Analog Computations

Wolfgang Maass, Pekka Orponen

https://doi.org/10.1162/089976698300017359 | pp. 1071-1095

We introduce a model for analog computation with discrete time in the presence of analog noise that is flexible enough to cover the most important concrete cases, such as noisy analog neural nets and networks of spiking neurons. This model subsumes the classical model for digital computation in the presence of noise. We show that the presence of arbitrarily small amounts of analog noise reduces the power of analog computational models to that of finite automata, and we also prove a new type of upper bound for the VC-dimension of computational models with analog noise.

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63. Category Learning Through Multimodality Sensing

Virginia R. de, Dana H. Ballard

https://doi.org/10.1162/089976698300017368 | pp. 1097-1117

Humans and other animals learn to form complex categories without receiving a target output, or teaching signal, with each input pattern. In contrast, most computer algorithms that emulate such performance assume the brain is provided with the correct output at the neuronal level or require grossly unphysiological methods of information propagation. Natural environments do not contain explicit labeling signals, but they do contain important information in the form of temporal correlations between sensations to different sensory modalities, and humans are affected by this correlational structure (Howells, 1944; McGurk & MacDonald, 1976; MacDonald & McGurk, 1978; Zellner & Kautz, 1990; Durgin & Proffitt, 1996). In this article we describe a simple, unsupervised neural network algorithm that also uses this natural structure. Using only the co-occurring patterns of lip motion and sound signals from a human speaker, the network learns separate visual and auditory speech classifiers that perform comparably to supervised networks.

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64. A Hierarchical Model of Binocular Rivalry

Peter Dayan

https://doi.org/10.1162/089976698300017377 | pp. 1119-1135

Binocular rivalry is the alternating percept that can result when the two eyes see different scenes. Recent psychophysical evidence supports the notion that some aspects of binocular rivalry bear functional similarities to other bistable percepts. We build a model based on the hypothesis (Logothetis & Schall, 1989; Leopold & Logothetis, 1996; Logothetis, Leopold & Sheinberg, 1996) that alternation can be generated by competition between top-down cortical explanations for the inputs, rather than by direct competition between

the inputs. Recent neurophysiological evidence shows that some binocular neurons are modulated with the changing percept; others are not, even if they are selective between the stimuli presented to the eyes. We extend our model to a hierarchy to address these effects.

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65. Efficient Learning in Boltzmann Machines Using Linear Response Theory

Hilbert J. Kappen, Francisco B. Rodr■■guez

https://doi.org/10.1162/089976698300017386 | pp. 1137-1156

The learning process in Boltzmann machines is computationally very expensive. The computational complexity of the exact algorithm is exponential in the number of neurons. We present a new approximate learning algorithm for Boltzmann machines, based on mean-field theory and the linear response theorem. The computational complexity of the algorithm is cubic in the number of neurons. In the absence of hidden units, we show how the weights can be directly computed from the fixed-point equation of the learning rules. Thus, in this case we do not need to use a gradient descent procedure for the learning process. We show that the solutions of this method are close to the optimal solutions and give a significant improvement when correlations play a significant role. Finally, we apply the method to a pattern completion task and show good performance for networks up to 100 neurons.

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66. A Learning Theorem for Networks at Detailed Stochastic Equilibrium

Javier R. Movellan

https://doi.org/10.1162/089976698300017395 | pp. 1157-1178

This article analyzes learning in continuous stochastic neural networks defined by stochastic differential equations (SDE). In particular, it studies gradient descent learning rules to train the equilibrium solutions of these networks. A theorem is given that specifies sufficient conditions for the gradient descent learning rules to be local covariance statistics between two random variables: (1) an evaluator that is the same for all the network parameters and (2) a system variable that is independent of the learning objective. While this article focuses on continuous stochastic neural networks, the theorem applies to any other system with Boltzmann-like equilibrium distributions. The generality of the theorem suggests that instead of suppressing noise present in physical devices, a natural alternative is to use it to simplify the credit assignment problem. In deterministic networks, credit assignment requires an evaluation signal that is different for each node in the network. Surprisingly, when noise is not suppressed, all that is needed is an evaluator that is the same for the entire network and a local Hebbian signal. This modularization of signals greatly simplifies hardware and software implementations. The article shows how the theorem applies to four different learning objectives that span supervised, reinforcement, and unsupervised problems: (1) regression, (2) density estimation, (3) risk minimization, and (4) information maximization. Simulations, implementation issues, and implications for computational neuroscience are discussed.

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67. Asymmetric Dynamics in Optimal Variance Adaptation

Michael R. DeWeese, Anthony M. Zador

https://doi.org/10.1162/089976698300017403 | pp. 1179-1202

It has long been recognized that sensory systems adapt to their inputs. Here we formulate the problem of optimal variance estimation for a broad class of nonstationary signals. We show that under weak assumptions, the Bayesian optimal causal variance estimate shows asymmetric dynamics: an abrupt increase in variance is more readily detectable than an abrupt decrease. By contrast, optimal adaptation to the mean displays symmetric dynamics when the variance is held fixed. After providing several empirical examples and a simple intuitive argument for our main result, we prove that optimal adaptation is asymmetrical in a broad class of model environments. This observation makes specific and falsifiable predictions about the time course of adaptation in neurons probed with certain stimulus ensembles.

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68. Computation with Infinite Neural Networks

Christopher K. I. Williams

https://doi.org/10.1162/089976698300017412 | pp. 1203-1216

For neural networks with a wide class of weight priors, it can be shown that in the limit of an infinite number of hidden units, the prior over functions tends to a gaussian process. In this article, analytic forms are derived for the covariance function of the gaussian processes corresponding to networks with sigmoidal and gaussian hidden units. This allows predictions to be made efficiently using networks with an infinite number of hidden units and shows, somewhat paradoxically, that it may be easier to carry out Bayesian prediction with infinite networks rather than finite ones.

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69. Bayesian Radial Basis Functions of Variable Dimension

Chris Holmes, Bani K. Mallick

https://doi.org/10.1162/089976698300017421 | pp. 1217-1233

A Bayesian framework for the analysis of radial basis functions (RBF) is proposed that accommodates uncertainty in the dimension of the model. A distribution is defined over the space of all RBF models of a given basis function, and posterior densities are computed using reversible jump Markov chain Monte Carlo samplers (Green, 1995). This alleviates the need to select the architecture during the modeling process. The resulting networks are shown to adjust their size to the complexity of the data.

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70. Absence of Cycles in Symmetric Neural Networks

Xin Wang, Arun Jagota, Fernanda Botelho, Max Garzón

https://doi.org/10.1162/089976698300017430 | pp. 1235-1249

For a given recurrent neural network, a discrete-time model may have asymptotic dynamics different from the one of a related continuous-time model. In this article, we consider a discrete-time model that discretizes the continuous-time leaky integrat or model and study its parallel, sequential, block-sequential, and distributed dynamics for symmetric networks. We provide sufficient (and in many cases necessary) conditions for the discretized model to have the same cycle-free dynamics of the corresponding continuous-time model in symmetric networks.

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71. Pattern Generation by Two Coupled Time-Discrete Neural Networks with Synaptic Depression

Walter Senn, Th. Wannier, J. Kleinle, H. R. Lüscher, Lorenz K. Müller, Jürg Streit, Kuno Wyler

https://doi.org/10.1162/089976698300017449 | pp. 1251-1275

Numerous animal behaviors, such as locomotion in vertebrates, are produced by rhythmic contractions that alternate between two muscle groups. The neuronal networks generating such alternate rhythmic activity are generally thought to rely on pacemaker cells or well-designed circuits consisting of inhibitory and excitatory neurons. However, experiments in organotypic cultures of embryonic rat spinal cord have shown that neuronal networks with purely excitatory and random connections may oscillate due to their synaptic depression, even without pacemaker cells. In this theoretical study, we investigate what happens if two such networks are symmetrically coupled by a small number of excitatory connections. We discuss a time-discrete mean-field model describing the average activity and the average synaptic depression of the two networks. Depending on the parameter values of the depression, the oscillations will be in phase, antiphase, quasiperiodic, or phase trapped. We put forward the hypothesis that pattern generators may rely on activity-dependent tuning of synaptic depression.

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72. Computational Studies of Lateralization of Phoneme Sequence Generation

James A. Reggia, Sharon Goodall, Yuri Shkuro

https://doi.org/10.1162/089976698300017458 | pp. 1277-1297

The mechanisms underlying cerebral lateralization of language are poorly understood. Asymmetries in the size of hemispheric regions and other factors have been suggested as possible underlying causal factors, and the corpus callosum (interhemispheric connections) has also been postulated to play a role. To examine these issues, we created a neural model consisting of paired cerebral hemispheric regions interacting via the corpus callosum. The model was trained to generate the correct sequence of phonemes for 50 monosyllabic words (simulated reading aloud) under a variety of assumptions about hemispheric asymmetries and callosal effects. After training, the ability of the full model and each hemisphere acting alone to perform this task was measured. Lateralization occurred readily toward the side having larger size, higher excitability, or higher-learning-rate parameter. Lateralization appeared most readily and intensely with strongly inhibitory callosal connections, supporting past arguments that the effective functionality of the corpus callosum is inhibitory. Many of the results are interpretable as the outcome of a "race to learn" between the model's two hemispheric regions, leading to the concept that asymmetric hemispheric plasticity is a critical common causative factor in lateralization. To our knowledge, this is the first computational model to demonstrate spontaneous lateralization of function, and it suggests that such models can be useful for understanding the mechanisms of cerebral lateralization.

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73. Nonlinear Component Analysis as a Kernel Eigenvalue Problem

Bernhard Schölkopf, Alexander J. Smola, Klaus■Robert Müller

https://doi.org/10.1162/089976698300017467 | pp. 1299-1319

A new method for performing a nonlinear form of principal component analysis is proposed. By the use of integral operator kernel functions, one can efficiently compute principal components in high-dimensional feature spaces, related to input space by some nonlinear map—for instance, the space of all possible five-pixel products in 16 × 16 images. We give the derivation of the method and present experimental results on polynomial feature extraction for pattern recognition.

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74. Chaotic Balanced State in a Model of Cortical Circuits

Carl van Vreeswijk, Haim Sompolinsky

https://doi.org/10.1162/089976698300017214 | pp. 1321-1371

The nature and origin of the temporal irregularity in the electrical activity of cortical neurons in vivo are not well understood. We consider the hypothesis that this irregularity is due to a balance of excitatory and inhibitory currents into the cortical cells. We study a network model with excitatory and inhibitory populations of simple binary units. The internal feedback is mediated by relatively large synaptic strengths, so that the magnitude of the total excitatory and inhibitory feedback is much larger than the neuronal threshold. The connectivity is random and sparse. The mean number of connections per unit is large, though small compared to the total number of cells in the network. The network also receives a large, temporally regular input from external sources. We present an analytical solution of the mean-field theory of this model, which is exact in the limit of large network size. This theory reveals a new cooperative stationary state of large networks, which we term a balanced state. In this state, a balance between the excitatory and inhibitory inputs emerges dynamically for a wide range of parameters, resulting in a net input whose temporal fluctuations are of the same order as its mean. The internal synaptic inputs act as a strong negative feedback, which linearizes the population responses to the external drive despite the strong nonlinearity of the individual cells. This feedback also greatly stabilizes the system's state and enables it to track a time-dependent input on time scales much shorter than the time constant of a single cell. The spatiotemporal statistics of the balanced state are calculated. It is shown that the autocorrelations decay on a short time scale, yielding an approximate Poissonian temporal statistics. The activity levels of single cells are broadly distributed, and their distribution exhibits a skewed shape with a long power-law tail. The chaotic nature of the balanced state is revealed by showing that the evolution of the microscopic state of the network is extremely sensitive to small deviations in its initial conditions. The balanced state generated by the sparse, strong connections is an asynchronous chaotic state. It is accompanied by weak spatial cross-correlations, the strength of which vanishes in the limit of large network size. This is in contrast to the synchronized chaotic states exhibited by more conventional network models with high connectivity of weak synapses.

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75. Blind Source Separation and Deconvolution: The Dynamic Component Analysis Algorithm

Hagai Attias, Christoph E. Schreiner

https://doi.org/10.1162/089976698300017223 | pp. 1373-1424

We derive a novel family of unsupervised learning algorithms for blind separation of mixed and convolved sources. Our approach is based on formulating the separation problem as a learning task of a spatiotemporal generative model, whose parameters are adapted iteratively to minimize suitable error

functions, thus ensuring stability of the algorithms. The resulting learning rules achieve separation by exploiting high-order spatiotemporal statistics of the mixture data. Different rules are obtained by learning generative models in the frequency and time domains, whereas a hybrid frequency-time model leads to the best performance. These algorithms generalize independent component analysis to the case of convolutive mixtures and exhibit superior performance on instantaneous mixtures. An extension of the relative-gradient concept to the spatiotemporal case leads to fast and efficient learning rules with equivariant properties. Our approach can incorporate information about the mixing situation when available, resulting in a "semiblind" separation method. The spatiotemporal redundancy reduction performed by our algorithms is shown to be equivalent to information-rate maximization through a simple network. We illustrate the performance of these algorithms by successfully separating instantaneous and convolutive mixtures of speech and noise signals.

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76. Blind Source Separation and Deconvolution: The Dynamic Component Analysis Algorithm

Hagai Attias, Christoph E. Schreiner

https://doi.org/10.1162/neco.1998.10.6.1373 | pp. 1373-1424

We derive a novel family of unsupervised learning algorithms for blind separation of mixed and convolved sources. Our approach is based on formulating the separation problem as a learning task of a spatiotemporal generative model, whose parameters are adapted iteratively to minimize suitable error functions, thus ensuring stability of the algorithms. The resulting learning rules achieve separation by exploiting high-order spatiotemporal statistics of the mixture data. Different rules are obtained by learning generative models in the frequency and time domains, whereas a hybrid frequency-time model leads to the best performance. These algorithms generalize independent component analysis to the case of convolutive mixtures and exhibit superior performance on instantaneous mixtures. An extension of the relative-gradient concept to the spatiotemporal case leads to fast and efficient learning rules with equivariant properties. Our approach can incorporate information about the mixing situation when available, resulting in a “semiblind” separation method. The spatiotemporal redundancy reduction performed by our algorithms is shown to be equivalent to information-rate maximization through a simple network. We illustrate the performance of these algorithms by successfully separating instantaneous and convolutive mixtures of speech and noise signals.

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77. Bias/Variance Decompositions for Likelihood-Based Estimators

Tom Heskes

https://doi.org/10.1162/089976698300017232 | pp. 1425-1433

The bias/variance decomposition of mean-squared error is well understood and relatively straightforward. In this note, a similar simple decomposition is derived, valid for any kind of error measure that, when using the appropriate probability model, can be derived from a Kullback-Leibler divergence or log-likelihood.

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78. The Influence Function of Principal Component Analysis by Self-Organizing Rule

Isao Higuchi, Shinto Eguchi

https://doi.org/10.1162/089976698300017241 | pp. 1435-1444

This article is concerned with a neural network approach to principal component analysis (PCA). An algorithm for PCA by the self-organizing rule has been proposed and its robustness observed through the simulation study by Xu and Yuille (1995). In this article, the robustness of the algorithm against outliers is investigated by using the theory of influence function. The influence function of the principal component vector is given in an explicit form. Through this expression, the method is shown to be robust against any directions orthogonal to the principal component vector. In addition, a statistic generated by the self-organizing rule is proposed to assess the influence of data in PCA.

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79. A Sparse Representation for Function Approximation

Tomaso Poggio, Federico Girosi

https://doi.org/10.1162/089976698300017250 | pp. 1445-1454

We derive a new general representation for a function as a linear combination of local correlation kernels at optimal sparse locations (and scales) and characterize its relation to principal component analysis, regularization, sparsity principles, and support vector machines.

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80. An Equivalence Between Sparse Approximation and Support Vector Machines

Federico Girosi

https://doi.org/10.1162/089976698300017269 | pp. 1455-1480

This article shows a relationship between two different approximation techniques: the support vector machines (SVM), proposed by V. Vapnik (1995) and a sparse approximation scheme that resembles the basis pursuit denoising algorithm (Chen, 1995; Chen, Donoho, and Saunders, 1995). SVM is a technique that can be derived from the structural risk minimization principle (Vapnik, 1982) and can be used to estimate the parameters of several different approximation schemes, including radial basis functions, algebraic and trigonometric polynomials, B-splines, and some forms of multilayer perceptrons. Basis pursuit denoising is a sparse approximation technique in which a function is reconstructed by using a small number of basis functions chosen from a large set (the dictionary). We show that if the data are noiseless, the modified version of basis pursuit denoising proposed in this article is equivalent to SVM in the following sense: if applied to the same data set, the two techniques give the same solution, which is obtained by solving the same quadratic programming problem. In the appendix, we present a derivation of the SVM technique in one framework of regularization theory, rather than statistical learning theory, establishing a connection between SVM, sparse approximation, and regularization theory.

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81. Extended Kalman Filter–Based Pruning Method for Recurrent Neural Networks

John Sum, Laiwan Chan, Chi-Sing Leung, Gilbert H. Young
https://doi.org/10.1162/089976698300017278 | pp. 1481-1505

Pruning is one of the effective techniques for improving the generalization error of neural networks. Existing pruning techniques are derived mainly from the viewpoint of energy minimization, which is commonly used in gradient-based learning methods. In recurrent networks, extended Kalman filter (EKF)-based training has been shown to be superior to gradient-based learning methods in terms of speed. This article explains a pruning procedure for recurrent neural networks using EKF training. The sensitivity of a posterior probability is used as a measure of the importance of a weight instead of error sensitivity since posterior probability density is readily obtained from this training method. The pruning procedure is tested using three problems: (1) the prediction of a simple linear time series, (2) the identification of a nonlinear system, and (3) the prediction of an exchange-rate time series. Simulation results demonstrate that the proposed pruning method is able to reduce the number of parameters and improve the generalization ability of a recurrent network.

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82. Transform-Invariant Recognition by Association in a Recurrent Network

Néstor Parga, Edmund T. Rolls
https://doi.org/10.1162/089976698300017287 | pp. 1507-1525

Objects can be recognized independently of the view they present, of their position on the retina, or their scale. It has been suggested that one basic mechanism that makes this possible is a memory effect, or a trace, that allows associations to be made between consecutive views of one object. In this work, we explore the possibility that this memory trace is provided by the sustained activity of neurons in layers of the visual pathway produced by an extensive recurrent connectivity. We describe a model that contains this high recurrent connectivity and synaptic efficacies built with contributions from associations between pairs of views that is simple enough to be treated analytically. The main result is that there is a change of behavior as the strength of the association between views of the same object, relative to the association within each view of an object, increases. When its value is small, sustained activity in the network is produced by the views themselves. As it increases above a threshold value, the network always reaches a particular state (which represents the object) independent of the particular view that was seen as a stimulus. In this regime, the network can still store an extensive number of objects, each defined by a finite (although it can be large) number of views.

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83. Retrieval Dynamics in Oscillator Neural Networks

Toshio Aoyagi, Katsunori Kitano
https://doi.org/10.1162/089976698300017296 | pp. 1527-1546

We present an analytical approach that allows us to treat the long-time behavior of the recalling process in an oscillator neural network. It is well known that in coupled oscillatory neuronal systems, under suitable conditions, the original dynamics can be reduced to a simpler phase dynamics. In this description, the phases of the oscillators can be regarded as the timings of the neuronal spikes. To attempt an analytical treatment of the recalling dynamics of such a system, we study a simplified model in which we discretize time and assume a synchronous updating rule. The theoretical results show that the retrieval dynamics is described by recursion equations for some macroscopic parameters, such as an overlap with the retrieval

pattern. We then treat the noise components in the local field, which arise from the learning of the unretrieved patterns, as gaussian variables. However, we take account of the temporal correlation between these noise components at different times. In particular, we find that this correlation is essential for correctly predicting the behavior of the retrieval process in the case of autoassociative memory. From the derived equations, the maximal storage capacity and the basin of attraction are calculated and graphically displayed. We also consider the more general case that the network retrieves an ordered sequence of phase patterns. In both cases, the basin of attraction remains sufficiently wide to recall the memorized pattern from a noisy one, even near saturation. The validity of these theoretical results is supported by numerical simulations. We believe that this model serves as a convenient starting point for the theoretical study of retrieval dynamics in general oscillatory systems.

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84. A Fast and Robust Cluster Update Algorithm for Image Segmentation in Spin-Lattice Models Without Annealing—Visual Latencies Revisited

Ralf Opara, Florentin Wörgötter

https://doi.org/10.1162/089976698300017304 | pp. 1547-1566

Image segmentation in spin-lattice models relies on the fast and reliable assignment of correct labels to those groups of spins that represent the same object. Commonly used local spin-update algorithms are slow because in each iteration only a single spin is flipped and a careful annealing schedule has to be designed in order to avoid local minima and correctly label larger areas. Updating of complete spin clusters is more efficient, but often clusters that should represent different objects will be conjoined. In this study, we propose a cluster update algorithm that, similar to most local update algorithms, calculates an energy function and determines the probability for flipping a whole cluster of spins by the energy gain calculated for a neighborhood of the regarded cluster. The novel algorithm, called energy-based cluster update (ECU algorithm), is compared to its predecessors. A convergence proof is derived, and it is shown that the algorithm outperforms local update algorithms by far in speed and reliability. At the same time it is more robust and noise tolerant than other versions of cluster update algorithms, making annealing completely unnecessary. The reduction in computational effort achieved this way allows us to segment real images in about 1-5 sec on a regular workstation. The ECU-algorithm can recover fine details of the images, and it is to a large degree robust with respect to luminance-gradients across objects. In a final step, we introduce luminance dependent visual latencies (Opara and Worgotter, 1996; Worgotter, Opara, Funke, and Eysel, 1996) into the spin-lattice model. This step guarantees that only spins representing pixels with similar luminance become activated at the same time. The energy function is then computed only for the interaction of the regarded cluster with the currently active spins. This latency mechanism improves the quality of the image segmentation by another 40%. The results shown are based on the evaluation of gray-level differences. It is important to realize that all algorithmic components can be transferred easily to arbitrary image features, like disparity, texture, and motion.

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85. Probability Density Methods for Smooth Function Approximation and Learning in Populations of Tuned Spiking Neurons

Terence D. Sanger

https://doi.org/10.1162/089976698300017313 | pp. 1567-1586

This article proposes a new method for interpreting computations performed by populations of spiking neurons. Neural firing is modeled as a rate-modulated random process for which the behavior of a neuron in response to external input can be completely described by its tuning function. I show that under certain conditions, cells with any desired tuning functions can be approximated using only spike coincidence detectors and linear operations on the spike output of existing cells. I show examples of adaptive algorithms based on only spike data that cause the underlying cell-tuning curves to converge according to standard supervised and unsupervised learning algorithms. Unsupervised learning based on principal components analysis leads to independent cell spike trains. These results suggest a duality relationship between the random discrete behavior of spiking cells and the deterministic smooth behavior of their tuning functions. Classical neural network approximation methods and learning algorithms based on continuous variables can thus be implemented within networks of spiking neurons without the need to make numerical estimates of the intermediate cell firing rates.

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86. A Potts Neuron Approach to Communication Routing

Jari Häkkinen, M. Lagerholm, Carsten Peterson, Bo Söderberg

https://doi.org/10.1162/089976698300017322 | pp. 1587-1599

A feedback neural network approach to communication routing problems is developed, with emphasis on multiple shortest path problems, with several requests for transmissions between distinct start and end nodes. The basic ingredients are a set of Potts neurons for each request, with interactions designed to minimize path lengths and prevent overloading of network arcs. The topological nature of the problem is conveniently handled using a propagator matrix approach. Although the constraints are global, the algorithmic steps are based entirely on local information, facilitating distributed implementations. In the polynomially solvable single-request case, the approach reduces to a fuzzy version of the Bellman-Ford algorithm. The method is evaluated for synthetic problems of varying sizes and load levels, by comparing to exact solutions from a branch-and-bound method, or to approximate solutions from a simple heuristic. With very few exceptions, the Potts approach gives high-quality legal solutions. The computational demand scales merely as the product of the numbers of requests, nodes, and arcs.

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87. Analog Versus Digital: Extrapolating from Electronics to Neurobiology

Rahul Sarpeshkar

https://doi.org/10.1162/089976698300017052 | pp. 1601-1638

We review the pros and cons of analog and digital computation. We propose that computation that is most efficient in its use of resources is neither analog computation nor digital computation but, rather, a mixture of the two forms. For maximum efficiency, the information and information-processing resources of the hybrid form must be distributed over many wires, with an optimal signal-to-noise ratio per wire. Our results suggest that it is likely that the brain computes in a hybrid fashion and that an underappreciated and important reason for the efficiency of the human brain, which consumes only 12 W, is the hybrid and distributed nature of its architecture.

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88. Employing the Z-Transform to Optimize the Calculation of the Synaptic Conductance of NMDA and Other Synaptic Channels in Network Simulations

J. Köhn, Florentin Wörgötter

https://doi.org/10.1162/089976698300017061 | pp. 1639-1651

Calculation of the total conductance change induced by multiple synapses at a given membrane compartment remains one of the most time-consuming processes in biophysically realistic neural network simulations. Here we show that this calculation can be achieved in a highly efficient way even for multiply converging synapses with different delays by means of the zeta-transform. Using the example of an NMDA synapse, we show that every update of the total conductance is achieved by an iterative process requiring at most three recent multiplications, which together need only the history values from the two most recent iterations. A major advantage is that this small computational load is independent of the number of synapses simulated. A benchmark comparison to other techniques demonstrates superior performance of the zeta-transform. Nonvoltage-dependent synaptic channels can be treated similarly (Olshausen, 1990; Brettle & Niebur, 1994), and the technique can also be generalized to other synaptic channels.

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89. Site-Selective Autophosphorylation of Ca²⁺/Calmodulin-Dependent Protein Kinase II as a Synaptic Encoding Mechanism

Christopher J. Coomber

https://doi.org/10.1162/089976698300017070 | pp. 1653-1678

A detailed kinetic model of the Ca2+/calmodulin-dependent protein kinase II (CaMKII) is presented in which subunits undergo autophosphorylation at several sites in a manner that depends on the frequency and duration of Ca2+ spikes. It is shown that high-frequency stimulation causes autophosphorylation of the autonomy site (Thr286), and promotes persistent catalytic activity. On the other hand, low-frequency stimulation is shown to cause autophosphorylation of an inhibitory site (Thr306), which prevents subunit activation. This site-selective autophosphorylation provides the basis for a molecular switch. When activated by a strong stimulus, the switch remains on for many minutes, even in the presence of a CaMKII-specific phosphatase. However, prolonged low-frequency stimulation disables the switch, and influences the response to subsequent stimulation. It is conceivable that a regulatory mechanism such as this may permit CaMKII to mediate synaptic frequency encoding and thereby direct an appropriate change in synaptic efficacy. It is indicated how the behavior of the model may relate to the induction of long-term potentiation.

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90. Ion Channel Stochasticity May Be Critical in Determining the Reliability and Precision of Spike Timing

Elad Schneidman, Barry Freedman, Idan Segev

https://doi.org/10.1162/089976698300017089 | pp. 1679-1703

The firing reliability and precision of an isopotential membrane patch consisting of a realistically large number of ion channels is investigated using a stochastic Hodgkin-Huxley (HH) model. In sharp contrast to the deterministic HH model, the biophysically inspired stochastic model reproduces qualitatively the different

reliability and precision characteristics of spike firing in response to DC and fluctuating current input in neocortical neurons, as reported by Mainen & Sejnowski (1995). For DC inputs, spike timing is highly unreliable; the reliability and precision are significantly increased for fluctuating current input. This behavior is critically determined by the relatively small number of excitable channels that are opened near threshold for spike firing rather than by the total number of channels that exist in the membrane patch. Channel fluctuations, together with the inherent bistability in the HH equations, give rise to three additional experimentally observed phenomena: subthreshold oscillations in the membrane voltage for DC input, "spontaneous" spikes for subthreshold inputs, and "missing" spikes for suprathreshold inputs. We suggest that the noise inherent in the operation of ion channels enables neurons to act as "smart" encoders. Slowly varying, uncorrelated inputs are coded with low reliability and accuracy and, hence, the information about such inputs is encoded almost exclusively by the spike rate. On the other hand, correlated presynaptic activity produces sharp fluctuations in the input to the postsynaptic cell, which are then encoded with high reliability and accuracy. In this case, information about the input exists in the exact timing of the spikes. We conclude that channel stochasticity should be considered in realistic models of neurons.

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91. Fast Temporal Encoding and Decoding with Spiking Neurons

D. Horn, Sharon Levanda

https://doi.org/10.1162/089976698300017098 | pp. 1705-1720

We propose a simple theoretical structure of interacting integrate-and-fire neurons that can handle fast information processing and may account for the fact that only a few neuronal spikes suffice to transmit information in the brain. Using integrate-and-fire neurons that are subjected to individual noise and to a common external input, we calculate their first passage time (FPT), or interspike interval. We suggest using a population average for evaluating the FPT that represents the desired information. Instantaneous lateral excitation among these neurons helps the analysis. By employing a second layer of neurons with variable connections to the first layer, we represent the strength of the input by the number of output neurons that fire, thus decoding the temporal information. Such a model can easily lead to a logarithmic relation as in Weber's law. The latter follows naturally from information maximization if the input strength is statistically distributed according to an approximate inverse law.

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92. Linearization of F-I Curves by Adaptation

Bard Ermentrout

https://doi.org/10.1162/089976698300017106 | pp. 1721-1729

We show that negative feedback to highly nonlinear frequency-current (F-I) curves results in an effective linearization. (By highly nonlinear we mean that the slope at threshold is infinite or very steep.) We then apply this to a specific model for spiking neurons and show that the details of the adaptation mechanism do not affect the results. The crucial points are that the adaptation is slow compared to other processes and the unadapted F-I curve is highly nonlinear.

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93. Mutual Information, Fisher Information, and Population Coding

Nicolas Brunel, Jean■Pierre Nadal
https://doi.org/10.1162/089976698300017115 | pp. 1731-1757

In the context of parameter estimation and model selection, it is only quite recently that a direct link between the Fisher information and information-theoretic quantities has been exhibited. We give an interpretation of this link within the standard framework of information theory. We show that in the context of population coding, the mutual information between the activity of a large array of neurons and a stimulus to which the neurons are tuned is naturally related to the Fisher information. In the light of this result, we consider the optimization of the tuning curves parameters in the case of neurons responding to a stimulus represented by an angular variable.

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94. Synaptic Pruning in Development: A Computational Account

Gal Chechik, Isaac Meilijson, Eytan Ruppin
https://doi.org/10.1162/089976698300017124 | pp. 1759-1777

Research with humans and primates shows that the developmental course of the brain involves synaptic overgrowth followed by marked selective pruning. Previous explanations have suggested that this intriguing, seemingly wasteful phenomenon is utilized to remove, "erroneous" synapses. We prove that this interpretation is wrong if synapses are Hebbian. Under limited metabolic energy resources restricting the amount and strength of synapses, we show that memory performance is maximized if synapses are first overgrown and then pruned following optimal "minimal-value" deletion. This optimal strategy leads to interesting insights concerning childhood amnesia.

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95. Spatial Decorrelation in Orientation-Selective Cortical Cells

Alexander G. Dimitrov, Jack D. Cowan
https://doi.org/10.1162/089976698300017133 | pp. 1779-1795

We propose a model for the lateral connectivity of orientation-selective cells in the visual cortex. We study the properties of the input signal to the visual cortex and find new statistical structures that have not been processed in the retino-geniculate pathway. Using the idea that the system performs redundancy reduction of the incoming signals, we derive the lateral connectivity that will achieve this for a set of orientation-selective local circuits, as well as the complete spatial structure of a network composed of such circuits. We compare the results with various physiological measurements.

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96. Receptive Field Formation in Natural Scene Environments: Comparison of Single-Cell Learning Rules

Brian S. Blais, Nathan Intrator, Harel Z. Shouval, Leon N. Cooper
https://doi.org/10.1162/089976698300017142 | pp. 1797-1813

We study several statistically and biologically motivated learning rules using the same visual environment: one made up of natural scenes and the same single-cell neuronal architecture. This allows us to concentrate on the feature extraction and neuronal coding properties of these rules. Included in these rules are kurtosis

and skewness maximization, the quadratic form of the Bienenstock-Cooper-Munro (BCM) learning rule, and single-cell independent component analysis. Using a structure removal method, we demonstrate that receptive fields developed using these rules depend on a small portion of the distribution. We find that the quadratic form of the BCM rule behaves in a manner similar to a kurtosis maximization rule when the distribution contains kurtotic directions, although the BCM modification equations are computationally simpler.

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97. Neural Feature Abstraction from Judgments of Similarity

Michael Lee

https://doi.org/10.1162/089976698300017151 | pp. 1815-1830

The common neural network modeling practice of representing the elements of a task domain in terms of a set of features lacks justification if the features are derived through some form of ad hoc preabstraction. By examining a featural similarity model related to established multidimensional scaling techniques, a neural network is developed that generates features from similarity data and attaches weights to these features. The network performs a constrained search of a continuous solution space to determine the features and uses a previously developed regularization technique to minimize the number of features it derives. The network is demonstrated on artificial data, from which it recovers known features and weights, and on two real data sets involving the similarity of a set of geometric shapes and the abstract conceptual similarities of the 10 Arabic numerals. On the basis of these results, the relationship between the multidimensional scaling approach adopted by the network and an alternative additive clustering approach to feature extraction is discussed.

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98. Classification of Temporal Patterns in Dynamic Biological Networks

Patrick D. Roberts

https://doi.org/10.1162/089976698300017160 | pp. 1831-1846

A general method is presented to classify temporal patterns generated by rhythmic biological networks when synaptic connections and cellular properties are known. The method is discrete in nature and relies on algebraic properties of state transitions and graph theory. Elements of the set of rhythms generated by a network are compared using a metric that quantifies the functional differences among them. The rhythms are then classified according to their location in a metric space. Examples are given, and biological implications are discussed.

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99. Kernel-Based Equiprobabilistic Topographic Map Formation

Marc M. Van Hulle

https://doi.org/10.1162/089976698300017179 | pp. 1847-1871

We introduce a new unsupervised competitive learning rule, the kernel-based maximum entropy learning rule (kMER), which performs equiprobabilistic topographic map formation in regular, fixed-topology lattices, for use with nonparametric density estimation as well as nonparametric regression analysis. The receptive fields of the formal neurons are overlapping radially symmetric kernels, compatible with radial basis

functions (RBFs); but unlike other learning schemes, the radii of these kernels do not have to be chosen in an ad hoc manner: the radii are adapted to the local input density, together with the weight vectors that define the kernel centers, so as to produce maps of which the neurons have an equal probability to be active (equiprobabilistic maps). Both an "online" and a "batch" version of the learning rule are introduced, which are applied to nonparametric density estimation and regression, respectively. The application envisaged is blind source separation (BSS) from nonlinear, noisy mixtures.

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100. An Energy Function and Continuous Edit Process for Graph Matching

Andrew Finch, Richard C. Wilson, Edwin R. Hancock

https://doi.org/10.1162/089976698300017188 | pp. 1873-1894

The contributions of this article are twofold. First, we develop a new nonquadratic energy function for graph matching. The starting point is a recently reported mixture model that gauges relational consistency using a series of exponential functions of the Hamming distances between graph neighborhoods. We compute the effective neighborhood potentials associated with the mixture model by identifying the single probability function of zero Kullback divergence. This new energy function is simply a weighted sum of graph Hamming distances. The second contribution is to locate matches by graduated assignment. Rather than solving the mean-field saddle-point equations, which are intractable for our nonquadratic energy function, we apply the soft-assign ansatz to the derivatives of our energy function. Here we introduce a novel departure from the standard graduated assignment formulation of graph matching by allowing the connection strengths of the data graph to update themselves. The aim is to provide a means by which the structure of the data graph can be updated so as to rectify structural errors. The method is evaluated experimentally and is shown to outperform its quadratic counterpart.

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101. Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms

Thomas G. Dietterich

https://doi.org/10.1162/089976698300017197 | pp. 1895-1923

This article reviews five approximate statistical tests for determining whether one learning algorithm outperforms another on a particular learning task. These tests are compared experimentally to determine their probability of incorrectly detecting a difference when no difference exists (type I error). Two widely used statistical tests are shown to have high probability of type I error in certain situations and should never be used: a test for difference of two proportions and a paired-differences t test based on taking several random train-test splits. A third test, a paired-differences t test based on 10-fold cross-validation, exhibits somewhat elevated probability of type I error. A fourth test, McNemar's test, is shown to have low type I error. The fifth test is a new test, 5 x 2 cv, based on five iterations of twofold cross-validation. Experiments show that this test also has acceptable type I error. The article also measures the power (ability to detect algorithm differences when they do exist) of these tests. The cross-validated t test is the most powerful. The 5 x 2 cv test is shown to be slightly more powerful than McNemar's test. The choice of the best test is determined by the computational cost of running the learning algorithm. For algorithms that can be executed only once, McNemar's test is the only test with acceptable type I error. For algorithms that can be executed 10 times, the 5 x 2 cv test is recommended, because it is slightly more powerful and because it directly measures

variation due to the choice of training set.

102. Probability Density Estimation Using Entropy Maximization

Gad Miller, D. Horn

https://doi.org/10.1162/089976698300017205 | pp. 1925-1938

We propose a method for estimating probability density functions and conditional density functions by training on data produced by such distributions. The algorithm employs new stochastic variables that amount to coding of the input, using a principle of entropy maximization. It is shown to be closely related to the maximum likelihood approach. The encoding step of the algorithm provides an estimate of the probability distribution. The decoding step serves as a generative mode, producing an ensemble of data with the desired distribution. The algorithm is readily implemented by neural networks, using stochastic gradient ascent to achieve entropy maximization.

103. Competition for Neurotrophic Factors: Mathematical Analysis

Terry Elliott, N. R. Shadbolt

https://doi.org/10.1162/089976698300016927 | pp. 1939-1981

Neurotrophic factors, particularly the neurotrophin gene family of neurotrophic factors, are implicated in activity-dependent anatomical plasticity in the visual cortex and at the neuromuscular junction. Accumulating evidence implicates neurotrophic factors as possible mediators of activity-dependent competition between afferents, leading to the segregation of afferents' arbors on the target space. We present a biologically plausible mathematical model of competition for neurotrophic factors. We show that the model leads to anatomical segregation, provided that the levels of neurotrophic factors released in an activity-independent manner, or the levels available by exogenous infusion, are below a critical value, which we derive. Above this critical value, afferent segregation breaks down. We also show that the model segregates afferents even in the presence of very highly correlated patterns of afferent activity. The model is therefore ideally suited for application to the development of ocular dominance columns in the kitten visual cortex.

104. Why Does the Somatosensory Homunculus Have Hands Next to Face and Feet Next to Genitals? A Hypothesis

Martha J. Farah

https://doi.org/10.1162/089976698300016936 | pp. 1983-1985

November 15 1998 Why Does the Somatosensory Homunculus Have Hands Next to Face and Feet Next to Genitals? A Hypothesis In Special Collection: CogNet Martha J. Farah Martha J. Farah Department of Psychology, University of Pennsylvania, Philadelphia, PA 19104, U.S.A. Search for other works by this author on: This Site Google Scholar Author and Article Information Martha J. Farah Department of Psychology, University of Pennsylvania, Philadelphia, PA 19104, U.S.A. Received: August 22 1997 Accepted: April 16 1998 Online Issn: 1530-888X Print Issn: 0899-7667 © 1998 Massachusetts Institute of Technology1998 Neural Computation (1998) 10 (8): 1983–1985. https://doi.org/10.1162/089976698300016936 Article history Received: August 22 1997 Accepted: April 16

1998 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn MailTo Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Martha J. Farah; Why Does the Somatosensory Homunculus Have Hands Next to Face and Feet Next to Genitals? A Hypothesis. Neural Comput 1998; 10 (8): 1983–1985. doi: https://doi.org/10.1162/089976698300016936 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1998 Massachusetts Institute of Technology1998 Article PDF first page preview Close Modal You do not currently have access to this content.

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105. Extracting Oscillations: Neuronal Coincidence Detection with Noisy Periodic Spike Input

Richard Kempter, Wulfram Gerstner, J. Leo van Hemmen, Hermann Wagner

https://doi.org/10.1162/089976698300016945 | pp. 1987-2017

How does a neuron vary its mean output firing rate if the input changes from random to oscillatory coherent but noisy activity? What are the critical parameters of the neuronal dynamics and input statistics? To answer these questions, we investigate the coincidence-detection properties of an integrate-and-fire neuron. We derive an expression indicating how coincidence detection depends on neuronal parameters. Specifically, we show how coincidence detection depends on the shape of the postsynaptic response function, the number of synapses, and the input statistics, and we demonstrate that there is an optimal threshold. Our considerations can be used to predict from neuronal parameters whether and to what extent a neuron can act as a coincidence detector and thus can convert a temporal code into a rate code.

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106. Connecting Cortical and Behavioral Dynamics: Bimanual Coordination

Viktor Jirsa, A. Fuchs, J. A. Scott Kelso

https://doi.org/10.1162/089976698300016954 | pp. 2019-2045

For the paradigmatic case of bimanual coordination, we review levels of organization of behavioral dynamics and present a description in terms of modes of behavior. We briefly review a recently developed model of spatiotemporal brain activity that is based on short- and long-range connectivity of neural ensembles. This model is specified for the case of motor and sensorimotor units embedded in the neural sheet. Focusing on the cortical left-right symmetry, we derive a bimodal description of the brain activity that is connected to behavioral dynamics. We make predictions of global features of brain dynamics during coordination tasks and test these against experimental magnetoencephalogram (MEG) results. A key feature of our approach is that phenomenological laws at the behavioral level can be connected to a field-theoretical description of cortical dynamics.

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107. Constructive Incremental Learning from Only Local Information

Stefan Schaal, Christopher G. Atkeson

https://doi.org/10.1162/089976698300016963 | pp. 2047-2084

We introduce a constructive, incremental learning system for regression problems that models data by means of spatially localized linear models. In contrast to other approaches, the size and shape of the receptive field of each locally linear model, as well as the parameters of the locally linear model itself, are learned independently, that is, without the need for competition or any other kind of communication. Independent learning is accomplished by incrementally minimizing a weighted local cross-validation error. As a result, we obtain a learning system that can allocate resources as needed while dealing with the bias-variance dilemma in a principled way. The spatial localization of the linear models increases robustness toward negative interference. Our learning system can be interpreted as a nonparametric adaptive bandwidth smoother, as a mixture of experts where the experts are trained in isolation, and as a learning system that profits from combining independent expert knowledge on the same problem. This article illustrates the potential learning capabilities of purely local learning and offers an interesting and powerful approach to learning with receptive fields.

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108. Information Maximization and Independent Component Analysis: Is There a Difference?

Dragan Obradovi■, Gustavo Deco

https://doi.org/10.1162/089976698300016972 | pp. 2085-2101

This article provides a detailed and rigorous analysis of the two commonly used methods for redundancy reduction: linear independent component analysis (ICA) posed as a direct minimization of a suitably chosen redundancy measure and information maximization (InfoMax) of a continuous stochastic signal transmitted through an appropriate nonlinear network. The article shows analytically that ICA based on the Kullback-Leibler information as a redundancy measure and InfoMax lead to the same solution if the parameterization of the output nonlinear functions in the latter method is sufficiently rich. Furthermore, this work discusses the alternative redundancy measures not based on the Kullback-Leibler information distance. The practical issues of applying ICA and InfoMax are also discussed and illustrated on the problem of extracting statistically independent factors from a linear, pixel-by-pixel mixture of images.

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109. An Alternative Perspective on Adaptive Independent Component Analysis Algorithms

Mark Girolami

https://doi.org/10.1162/089976698300016981 | pp. 2103-2114

This article develops an extended independent component analysis algorithm for mixtures of arbitrary subgaussian and supergaussian sources. The gaussian mixture model of Pearson is employed in deriving a closed-form generic score function for strictly subgaussian sources. This is combined with the score function for a unimodal supergaussian density to provide a computationally simple yet powerful algorithm for performing independent component analysis on arbitrary mixtures of nongaussian sources.

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110. Density Estimation by Mixture Models with Smoothing Priors

Akio Utsugi

https://doi.org/10.1162/089976698300016990 | pp. 2115-2135

In the statistical approach for self-organizing maps (SOMs), learning is regarded as an estimation algorithm for a gaussian mixture model with a gaussian smoothing prior on the centroid parameters. The values of the hyperparameters and the topological structure are selected on the basis of a statistical principle. However, since the component selection probabilities are fixed to a common value, the centroids concentrate on areas with high data density. This deforms a coordinate system on an extracted manifold and makes smoothness evaluation for the manifold inaccurate. In this article, we study an extended SOM model whose component selection probabilities are variable. To stabilize the estimation, a smoothing prior on the component selection probabilities is introduced. An estimation algorithm for the parameters and the hyperparameters based on empirical Bayesian inference is obtained. The performance of density estimation by the new model and the SOM model is compared via simulation experiments.

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111. Complexity Issues in Natural Gradient Descent Method for Training Multilayer Perceptrons

Howard H. Yang, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976698300017007 | pp. 2137-2157

The natural gradient descent method is applied to train an n-m-1 multilayer perceptron. Based on an efficient scheme to represent the Fisher information matrix for an n-m-1 stochastic multilayer perceptron, a new algorithm is proposed to calculate the natural gradient without inverting the Fisher information matrix explicitly. When the input dimension n is much larger than the number of hidden neurons m, the time complexity of computing the natural gradient is O(n).

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112. Almost Linear VC-Dimension Bounds for Piecewise Polynomial Networks

Peter L. Bartlett, Vitaly Maiorov, Ron Meir

https://doi.org/10.1162/089976698300017016 | pp. 2159-2173

We compute upper and lower bounds on the VC dimension and pseudo-dimension of feedforward neural networks composed of piecewise polynomial activation functions. We show that if the number of layers is fixed, then the VC dimension and pseudo-dimension grow as WlogW, where W is the number of parameters in the network. This result stands in opposition to the case where the number of layers is unbounded, in which case the VC dimension and pseudo-dimension grow as W2. We combine our results with recently established approximation error rates and determine error bounds for the problem of regression estimation by piecewise polynomial networks with unbounded weights.

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113. The Diabolo Classifier

Holger Schwenk

https://doi.org/10.1162/089976698300017025 | pp. 2175-2200

We present a new classification architecture based on autoassociative neural networks that are used to learn discriminant models of each class. The proposed architecture has several interesting properties with respect to other model-based classifiers like nearest-neighbors or radial basis functions: it has a low computational complexity and uses a compact distributed representation of the models. The classifier is also well suited for the incorporation of a priori knowledge by means of a problem-specific distance measure. In particular, we will show that tangent distance (Simard, Le Cun, & Denker, 1993) can be used to achieve transformation invariance during learning and recognition. We demonstrate the application of this classifier to optical character recognition, where it has achieved state-of-the-art results on several reference databases. Relations to other models, in particular those based on principal component analysis, are also discussed.

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114. Online Learning from Finite Training Sets and Robustness to Input Bias

Peter Sollich, David Barber

https://doi.org/10.1162/089976698300017034 | pp. 2201-2217

We analyze online gradient descent learning from finite training sets at noninfinitesimal learning rates eta. Exact results are obtained for the time-dependent generalization error of a simple model system: a linear network with a large number of weights N, trained on p = alphaN examples. This allows us to study in detail the effects of finite training set size alpha on, for example, the optimal choice of learning rate eta. We also compare online and offline learning, for respective optimal settings of eta at given final learning time. Online learning turns out to be much more robust to input bias and actually outperforms offline learning when such bias is present; for unbiased inputs, online and offline learning perform almost equally well.

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115. Antipredictable Sequences: Harder to Predict Than Random Sequences

Huaiyu Zhu, Wolfgang Kinzel

https://doi.org/10.1162/089976698300017043 | pp. 2219-2230

For any discrete-state sequence prediction algorithm A, it is always possible, using an algorithm B no more complicated than A, to generate a sequence for which A's prediction is always wrong. For any prediction algorithm A and sequence x, there exists a sequence y no more complicated than x, such that if A performs better than random on x, then it will perform worse than random on y by the same margin. An example of a simple neural network predicting a bit sequence is used to illustrate this very general but not widely recognized phenomenon. This implies that any predictor with good performance must rely on some (usually implicitly) assumed prior distributions of the problem.